10.8 COBI: A Degree-of-56 Column-Bipartite Densely Connected Digital Ising Chip with 8b Spin Coefficients

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Abstract

We present a 65nm digital Ising chip with an advanced column-bipartite topology, featuring densely connected spins with a degree of 56 and 8b coefficients for mapping and solving computationally intensive combinatorial optimization problems. A fabricated Ising test chip with 64 spin processing elements with spin coefficients stored in flexible embedded SRAM demonstrates solving practical problems, such as MIMO detection for a wireless network, with a fast solution time (<72ns).

Combinatorial optimization problems (COPs) are prevalent in various industries, including logistics, networks, finance, drug discovery and wireless communications. Many of these problems are NP-hard and scale poorly on conventional CPUs and GPUs. Ising machines provide a promising alternative by mapping COP variables onto Ising spins, where spin interactions encode the energy function. The system then searches for low-energy states of the Ising Hamiltonian, typically using energy gradient descent combined with stochastic methods, such as annealing to escape local minima. Quantum annealers [1] have demonstrated the feasibility of large-scale Ising machines, but their applications are limited due to the extremely low operating temperature (<20mK) and high-power consumption (15-25kW). In contrast, CMOS Ising machines have emerged as viable approaches, achieving low power consumption and good scalability while operating at room temperature [2-6]. However, prior CMOS Ising machines suffer from poor problem-mapping capabilities, limiting their practical applications, due to fixed and low-degree hardware topologies, such as lattice and King's graphs [1-3]. A continuous-time Ising machine with embedded DRAM [4] implemented a King's graph topology with the degree of 8 (allowing up to 8 interconnects per spin) and 4b coefficients. An SRAM-based macro implemented a network of spins in an SRAM macro using an enhanced Chimera topology [5], inspired by a quantum annealer [1]. However, its precision is limited to ternary due to the limited spatial connectivity within the macro. A recent m-Zephyr topology [6] further advances connectivity and achieves a degree of 24 through cross-dimensional connections, yet it requires computationally intensive pre-processing for problem mapping. Due to the limited hardware topologies, prior works have primarily focused on showcasing simple benchmarks, such as Max-Cut problems, which can be mapped into any hardware topology regardless of connectivity and precision. In this work, we introduce a digital Ising chip with an advanced column-bipartite topology, featuring densely connected spins with a degree of 56 and 8b coefficients for mapping and solving practical problems, such as community and MIMO detection, with densely connected spins and high coefficient precisions. The digital Ising chip enhances reconfigurability by introducing flexibility in mapping spin coefficients to embedded memory. The fabricated 65nm test chip implements 64 densely connected spin processing elements (PEs) with 8b spin coefficients and demonstrates solving community and MIMO detection problems with a fast solution time (<72ns). Fig. 10.8.1 provides an overview of this work, highlighting its main design features with the operating principles of the Ising machine.

Figure 10.8.2 illustrates the overall architecture of a 65nm test chip, comprising 64 spin PEs arranged in a two-dimensional array, along with detailed circuit building block schematics and a spin layout. A PE integrates four modules: (i) a 512b 8T-SRAM for storing sixty-four 8b spin coefficient values; (ii) a near-memory MAC comprising eight multipliers, an adder tree, and an accumulator for accumulating gradients; (iii) a register for storing a binary spin state with a zero detector; (iv) a bus controller for orchestrating spin connectivity. Note that a MAC computes a gradient accumulation by multiplying eight 8b spin coefficients and the incoming eight binary spin states, adding them using an adder tree, and accumulating them. Then, the inverted sign bit of the accumulated gradient value is used to update the spin state. When the accumulated gradient value is zero, the spin state is inverted to introduce extra randomness, improving the chance of escaping local minima and reaching better sub-optimal solutions. A bus controller embedded in each spin PE plays a key role in implementing the column-bipartite topology by routing an updated spin state output to all other spins by connecting it to horizontal and vertical 8b spin buses distributed throughout the Ising core. In each cycle, all eight spin state outputs from a single column are connected to all other spins via buses. It takes eight clock cycles to complete a full cycle of spin interactions, thereby maximizing parallelism and achieving a significant speedup.

Figure 10.8.3 (top) shows the sequential activation of columns, where spins in an active column transmit their spin state outputs to vertical buses and spins in inactive columns receive incoming spin states from horizontal buses. Fig. 10.8.3 (bottom) illustrates detailed operating sequences of a spin in an activated and an inactive column. In each cycle, spins in an active column (in transmit or Tx mode) enable their bus controller switches to broadcast their spin state outputs to other spins via vertical and horizontal spin buses. Meanwhile, spins in inactive columns (in receive or Rx mode) compute interactions with eight spins in active columns by receiving their transmitted spin states and multiplying them with interaction coefficients retrieved from local 8T-SRAM via local bitlines (LBL). Eight element-wise multiplied results are then added (via an adder tree), accumulated, and then used to update a spin state at a spin register.

Figure 10.8.4 (top) shows measurement results of easy Max-Cut problems with a predefined ground state encoded as a clean monster image. Measured transient spin state maps (where white and black pixels indicate spin states +1 and -1) showcase fast convergence. It takes two cycles to reach convergence with the stabilized Ising Hamiltonian of -1792 at a ground state. Note that the top and bottom row transient spin state maps result from the same Max-Cut problem, sharing the same coefficients but with different initial spin states. Their converged spin maps appear inverted, but both represent the ground state with the minimum Ising Hamiltonian value. Fig. 10.8.4 (bottom) illustrates hard Max-Cut problems with random spin interaction coefficients, whose values range between -127 and +127. Measured results show that the chip achieves a solution time more than ~10^7× faster than the baseline CPU using the Metropolis algorithm. Statistics of the measured results (with 1000 samples) indicate that the Ising chip achieves lower mean Hamiltonians with less variation. Note that this work achieves better suboptimal results (with lower Hamiltonians), while requiring only eight clock cycles for convergence.

Figure 10.8.5 illustrates problem mappings and evaluation results for two computationally intensive problems with versatile and practical applications, requiring dense connections and high coefficient precision for solving them using the Ising chip. Each problem is mapped to the test chip with two different spin coefficient configurations, one with J coefficients only and the other with both J and h coefficients, where J and h indicate interaction and bias coefficients, respectively. Fig. 10.8.5 (top-right) shows the measured results for a 64-node community detection problem. A complex graph is embedded into the chip and initialized with random spin states. After the first run, spins form a clear bipartition. For the second run, we reprogram the J coefficient matrices by detaching all inter-community links by setting their corresponding J coefficients to zero, while keeping intra-community couplings. The chip then operates on each group of spins, bisecting them, resulting in four communities. Spin state map snapshots and their corresponding visualized graph views show clusters emerging and stabilizing through bisections. If more partitions are desired, we repeat the same process (reprogramming the J coefficients) as needed. Fig. 10.8.5 (bottom right) shows the measured results for a 24×24 BPSK MIMO detection problem for wireless communications. In the MIMO communication system, multiple antennas transmit and receive simultaneously, while a BPSK modulation constrains each transmitted symbol to one of the two possible data symbols. The MIMO detector's objective is to recover the original transmitted symbols from the received signals that are linearly mixed by the channel and corrupted by additive noise. Then, we directly map the MIMO detection problem to the Ising chip with the column-bipartite densely connected topology, where couplings between spins capture channel interactions and bias terms capture the influence of received data. A 24×24 BPSK MIMO detection problem is segmented and mapped to fit into the Ising chip with eight spins allocated for six active columns, while a column is allocated for h bias coefficients for active spins. Measured transient Hamiltonian traces show a faster time to solution than the baseline, which uses the CPU to run the Metropolis algorithm. Note that the column-bipartite topology is readily scalable for solving higher-order modulation (e.g., QPSK, 16-QAM), with the expectation of a significant reduction in latency, which is a critical performance bottleneck for next-generation wireless networks.

Figure 10.8.6 presents a comparison table that compares this work with the state of the art in digital and mixed-signal implementations. A die micrograph and a summary table of this work, fabricated with a 65nm process, are shown in Fig. 10.8.7. A spin PE occupies 120.75×62.5μm^2, and the measured power consumption is 518μW at 1.2V when solving hard Max-Cut problems with random coefficients.

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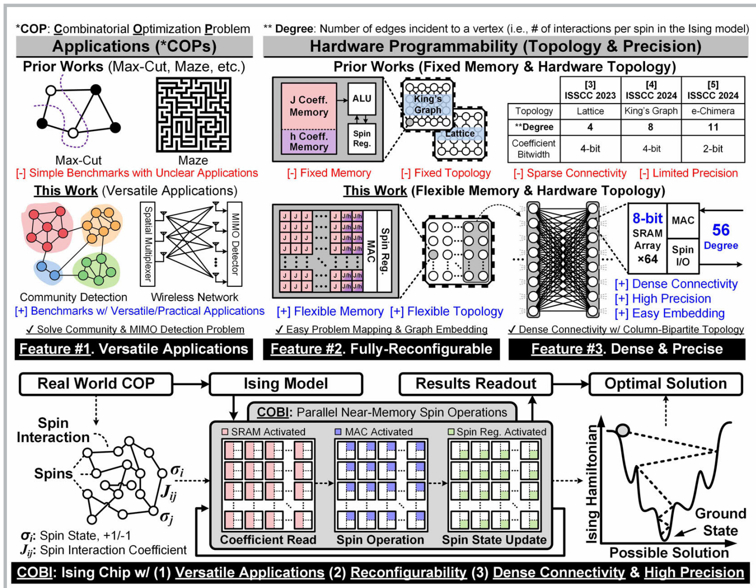


Figure 10.8.1: Introduction to COP problems, prior Ising works, and the fully digital Ising chip (COBI) with reconfigurability, dense connectivity, and high precision.

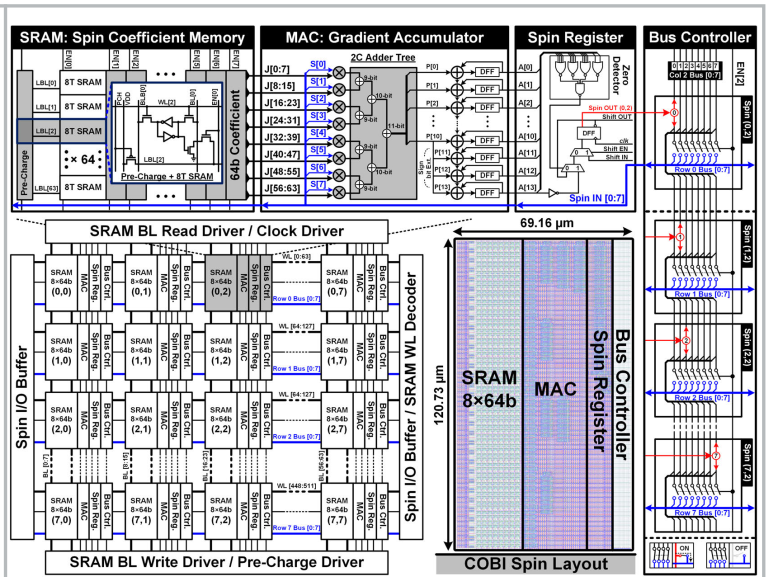


Figure 10.8.2: Overall architecture of the fabricated 65nm test chip, detailed spin structure, and a single spin layout.

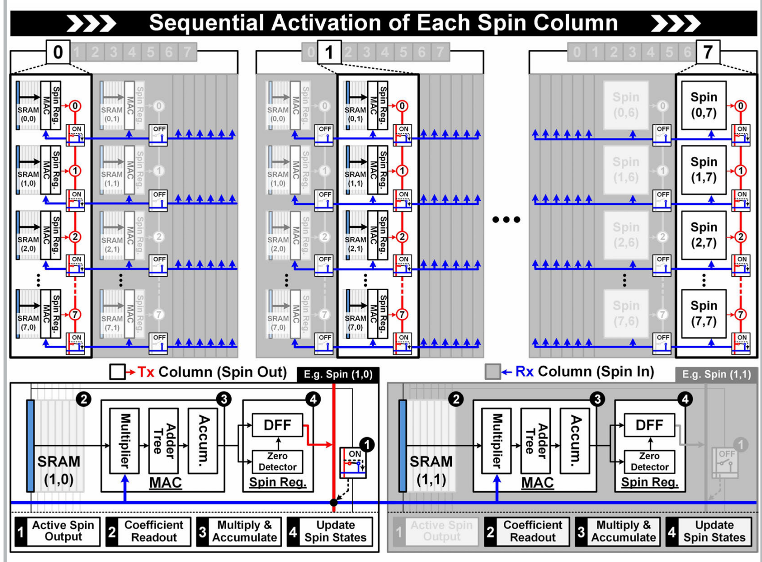


Figure 10.8.3: Sequential activation of spin column: Transmit (Tx) and receive (Rx) operations, coefficient readout, and MAC operation.

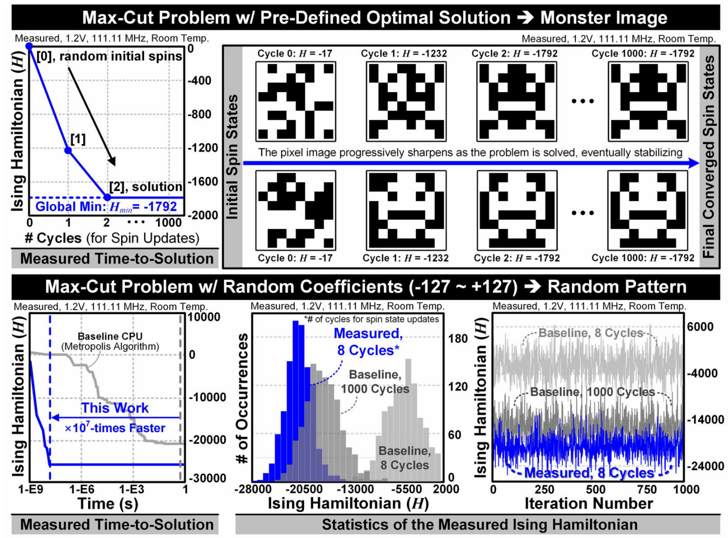


Figure 10.8.4: Measurement results of Max-Cut problems with transient Ising Hamiltonians (top-left) and transient spin maps (top-right), and hard Max-Cut problems with transient Ising Hamiltonians (bottom-left) and statistical distributions (bottom-right).

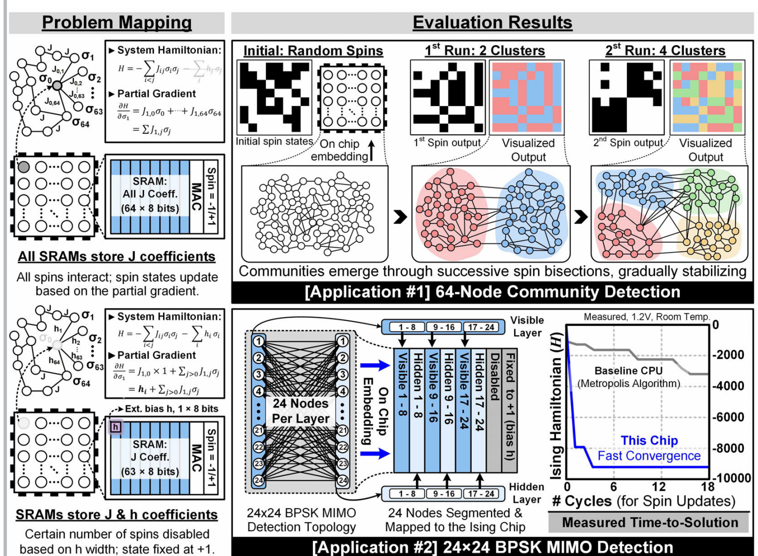
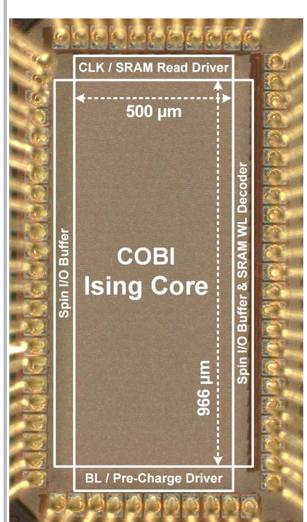


Figure 10.8.5: Mapping (left) and evaluated results (right) of 64-node community detection (top) and 24×24 BPSK MIMO detection (bottom) problems.

	JSSC 2022 [2]	ISSCC 2023 [3]	ISSCC 2024 [4]	ISSCC 2024 [5]	VLSI 2025 [6]	This Work
Technology (Circuit Type)	65nm CMOS (Mixed-Signal)	65nm CMOS (Analog)	65nm CMOS (Mixed-Signal)	65nm CMOS (Mixed-Signal)	65nm CMOS (Digital)	65nm CMOS (Digital)
Operating Temperature	Room Temp. (300K)	Room Temp. (300K)	Room Temp. (300K)	Room Temp. (300K)	Room Temp. (300K)	Room Temp. (300K)
Computing Type	Discrete-Time Mixed-Signal	Continuous-Time Analog	Continuous-Time Mixed-Signal	Continuous-Time Mixed-Signal	Discrete-Time Digital MAC	Discrete-Time Digital MAC
Memory-Centric Computing	Near-Memory Computing	Near-Memory Computing	Near-Memory Computing	Within-Memory Computing	Near-Memory Computing	Near-Memory Computing
Spin Memory	eDRAM Cell	CMOS Latch	eDRAM Cell	SRAM Cell	Register	Register
Spin State Representation	Digital (Binary State)	Analog (Latch Voltage)	Digital (Binary State)	Analog (Latch Voltage)	Digital (Binary State)	Digital (Binary State)
Spin Interaction	Mixed-Signal MAC	Latch Coupling Via Switches	Mixed-Signal MAC	Latch Coupling Via Switches	Digital MAC	Digital MAC
Graph Topology	King's Graph	Lattice	King's Graph	e-Chimera	m-Zephyr	Column Bipartite
# of Neighbors (Degree of Graphs)	8	4	8	11	24	56
Coeff. SRAM R/W During Operation	Required (via SRAM I/O)	Directly Read (No Write)	Required (via SRAM I/O)	Required (via SRAM I/O)	Directly Read (No Write)	Directly Read (No Write)
Coefficient Bitwidth	1-4bit	1bit	4bit	2bit	7bit	8bit
Supply Voltage	0.9-1.2V	0.75-1.05V	0.9-1.2V	0.8-1.4V	0.6-1.4V	0.8-1.4V
Tested Problems	Max-Cut Only					Max-Cut, MIMO, Community Detection
Time-to-Solution (TTS)	51200 ns (12800 Cycles)	<20ns	<20.7ns	<100 ns	<150ns (10 Cycles)	<72 ns* (8 Cycles)

*TTS of the proposed chip is measured at 1.2 V and 111.11 MHz while solving hard Max-Cut problems with all random coefficients

Figure 10.8.6: A comparison table with the state-of-the-art CMOS Ising machines.



Technology	65nm LP CMOS
Test Chip	Fully-Reconfigurable Digital Ising Chip
Core Circuit	64× Spin Processing Element Array
Topology (Degree)	Column-Bipartite Dense Connections 56 (# of interactions per spin)
Core/Spin Area	966×500 μm^2 / 120.75×62.5 μm^2
SRAM	64 × 512b (Store 64×64 8-bit Spin Coefficients)
Supply Voltage	Core: 0.8 V - 1.2 V, I/O: 3.3 V
Clock Freq.	111.11 MHz
Time-to-Solution (TTS)	<72 ns*
Measured Power Consumption	518 µW @ 1.2 V

*TTS of the proposed chip is measured at 1.2 V and 111.11 MHz while solving hard Max-Cut problems with all random coefficients

Figure 10.8.7: Die micrograph and a summary table.

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